

How Much of Truth Fits on a Single Axis?

Knowledge-Dependent Dimensionality and Unsupervised Polarity Recovery
in Large Language Model Representations

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Reference code: `truth_probe.py` — Model: Qwen2.5-1.5B

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Abstract

Two dominant families of methods read factual truth from the hidden states of language models. *Linear probes* project the residual stream onto a single learned or difference-of-means axis; *sparse autoencoders* (SAEs) expand the state against an overcomplete dictionary to disambiguate features in superposition. The first is cheap but treats truth as a static one-dimensional axis; the second is interpretable but costly. We study a third, deliberately minimal object: an *unsupervised* geometric probe obtained from the singular value decomposition (SVD) of paired true/false differences, readable at cost $O(d)$ per token with no training. On Qwen2.5-1.5B, under a held-out cross-validation protocol with a label-permutation null, we report three findings. (i) The unsupervised axis carries a *real* truth signal—held-out AUC up to 0.938 on clean minimal pairs, permutation $p = 0.005$ —but a full linear probe on the same raw states reaches 0.989; the gap is informative rather than incidental. (ii) That gap *scales with how well the model knows the fact*: ≈ 0.05 on facts the model plausibly knows, ≈ 0.20 on facts it plausibly does not. We read this as the truth representation being near one-dimensional when the fact is known, and diffusing across many directions when it is not. (iii) On mixed-polarity (affirmative/negated) data, the unsupervised SVD *recovers* the supervised polarity direction of Bürger et al. at cosine 0.959, without ever seeing the polarity label. We additionally test, and reject, a complex-plane “phase” extension: the apparent rotation of falsehoods reduces to the sign flip of the real axis, and the second dimension adds no separation on single-polarity data. We conclude that this probe is a cheap, training-free *structural* instrument—useful for monitoring and analysis—and is not a low-cost substitute for dictionary learning, which solves a different problem bounded by a capacity argument. Results are preliminary (single model, small datasets) and intended to be falsifiable.

1 Introduction

Large language models produce false statements with the same fluency as true ones. Predicting when a model is about to leave the truth—ideally during generation, without a second judge model—is a central reliability problem. Mechanistic interpretability has shown that many high-level properties, including the truth value of a factual statement, are encoded as linear directions in the residual stream [2, 4]. Two method families follow. The first is the *linear probe*: a classifier or difference-of-means axis whose projection separates true from false [1, 2, 3]. The second is *sparse dictionary learning*: project the state onto an overcomplete dictionary, potentially millions of atoms, to undo superposition and read monosemantic features [6, 7].

Each pays a price. A single static axis is fragile and low-capacity; a sparse autoencoder requires dedicated training and a large matrix multiply per token. This paper takes a third stance. Rather

than learning a probe or expanding a dictionary, we *read structure from geometry alone*, and ask a precise question: *how much of the truth structure does a single direction capture, and when does that capture fail?* We extract all hidden states in one forward pass, let an axis emerge without supervision from the SVD of paired true/false differences, and then—crucially—measure everything out of sample, with a permutation null that quantifies how much apparent accuracy is mere selection. The answer is modest but, we believe, clean: the axis reads truth cheaply when the fact is known, fails predictably when it is not, and—blind to the label—recovers the polarity structure that a prior supervised method constructed by hand.

Contributions.

1. An unsupervised, training-free truth probe at cost $O(d)$, with a rigorous held-out and permutation-tested evaluation that corrects the optimism of in-sample axis selection (Sec. 4.1).
2. Evidence that the *dimensionality of the truth representation is knowledge-dependent*: near one-dimensional for known facts, diffuse for unknown ones, quantified by the gap to a full linear probe (Sec. 4.2).
3. *Unsupervised recovery of the polarity direction t_P* of Bürger et al. [5]: the SVD of mixed-polarity differences aligns with the supervised t_P at cosine 0.959, with no polarity label (Sec. 4.4).
4. A negative result: a complex-plane / phase extension does not improve separation, and the “rotation” of falsehoods is the real-axis sign flip re-expressed in polar coordinates (Sec. 4.3).

2 Related work

Linear truth directions. [1] extract a knowledge direction without labels by enforcing logical consistency. [2] show truth has emergent linear structure and that difference-of-means probes generalize as well as more complex ones. [3] intervene along truthful directions at inference time; [4] frame representation reading top-down. Our axis shares the linear-direction hypothesis but is read from the geometry of paired differences rather than imposed by a trained probe.

A two-dimensional truth subspace. Closest to us, Bürger et al. [5] show that truth is represented not in a single direction but in a two-dimensional subspace spanned by a general truth direction t_G and a polarity-sensitive direction $t_P \sim \text{XOR}(\text{is_true}, \text{is_negated})$, and that probes trained on affirmative statements fail on negated ones. We confirm this structure in Qwen2.5-1.5B and show, as a methodological addition, that t_P can be recovered *without* polarity supervision.

Cross-layer dynamics. Recent monitors model the evolution of hidden states across depth rather than a single layer [9, 10]. Our depth trajectories are descriptive and consistent with this line.

Superposition and dictionaries. Superposition [8] motivates decomposing the residual stream into overcomplete dictionaries [6, 7]. This is the operation we explicitly do *not* perform; Sec. 5 argues why a single projection cannot substitute for it.

3 Method

3.1 Extraction and the unsupervised axis

We load Qwen2.5-1.5B frozen (28 blocks, hence 29 hidden-state levels: the embedding plus each block output, $d = 1536$) and, per sentence, run one forward pass with `output_hidden_states`, taking the last-token state at every layer. For N sentences this yields $H \in \mathbb{R}^{N \times 29 \times d}$.

The data are *minimal pairs*: each true sentence is matched to its false counterpart with the same subject and a flipped predicate (e.g. “The Sun is a star.” vs. “The Sun is a planet.”). Within a pair, the difference cancels the shared topic and leaves the truth signal. We form the intra-pair difference matrix D with rows $h_{\text{true}} - h_{\text{false}}$, and take its SVD,

$$D = USV^\top, \quad v_1 = V_1, \quad (1)$$

where v_1 is the dominant direction of the differences. The labels do not build v_1 ; the pairing does. A single orientation bit (“true is positive”), fit on training data, fixes the sign.

3.2 A falsifiable evaluation protocol

The central methodological point is that the axis, its calibration, the orientation bit, *and* the choice of peak layer must be fit on training data only. We therefore use k -fold cross-validation *over pairs* (a pair is never split across folds, which would leak the shared topic), fit everything on the training folds, and report held-out AUC averaged across folds. To quantify how much apparent accuracy is selection over the 29 layers, we run a *permutation null*: we randomly swap true/false *within* each pair (labels become noise, the pair structure is preserved) and recompute the maximum held-out AUC over layers; the p -value is the fraction of the null at or above the observed value. As a reference point we also fit an ℓ_2 -regularized logistic probe on the raw d -dimensional states, held out over the same pair folds.

3.3 The complex-plane extension we test

We additionally form a second axis v_2 , the SVD residual orthogonal to v_1 , treat the projection as $z = c_1 + i c_2$ with calibrated real and imaginary parts, and test whether magnitude- or phase-based scores improve separation over the real axis alone. They do not (Sec. 4.3); we keep the result because the negative finding is itself informative.

3.4 Polarity design

To test whether a genuine second dimension exists, we use a 2×2 design: each fact in four forms—affirmative-true (AT), affirmative-false (AF), negated-true (NT), negated-false (NF)—giving the XOR structure of [5]. We compute the affirmative and negated truth directions $t_A = \overline{AT} - \overline{AF}$, $t_N = \overline{NT} - \overline{NF}$, the general direction $t_G \propto t_A + t_N$, and the supervised polarity direction $t_P \propto t_A - t_N$. For the unsupervised recovery test, we run the difference SVD on minimal pairs that *mix* the two polarities, so the SVD never sees the polarity label, and compare its leading directions to the supervised t_P .

4 Results

All numbers are on Qwen2.5-1.5B with last-token pooling; AUC is out-of-fold unless stated.

Table 1: Held-out AUC (5-fold CV over pairs) at layer 16, Qwen2.5-1.5B. “1D” is the real axis; “MAG” the magnitude-gated polar score; “PHASE” the angular-deviation score; “2D” a logistic on (Re, Im); “Probe” an ℓ_2 logistic on raw states. The last column is the label-permutation p -value for the 1D axis.

Dataset	pairs	1D	MAG	PHASE	2D	Probe	p
Clean minimal pairs	45	0.938	0.936	0.928	0.931	0.989	0.005
CounterFact / TruthfulQA mix	250	0.638	0.640	0.676	0.651	0.841	0.005

4.1 A real but bounded truth signal

Table 1 reports the held-out evaluation at the peak layer (layer 16). On clean minimal pairs the unsupervised axis reaches AUC 0.938 with permutation $p = 0.005$ (null mean 0.588, 95th percentile 0.664): the geometric axis aligns with truth well beyond what layer selection alone can manufacture. The signal is therefore real. It is also *bounded*: a full linear probe on the same raw states reaches 0.989. The probe sees information the single axis discards.

4.2 The gap scales with model knowledge

The two rows of Table 1 differ not only in difficulty but in the *size of the probe gap*. On clean facts the model plausibly knows, the unsupervised axis (0.938) trails the full probe (0.989) by only ≈ 0.05 . On CounterFact—which contains many obscure relations a 1.5B model plausibly does not know (e.g. the mother tongue of a minor historical figure)—the axis (0.638) trails the probe (0.841) by ≈ 0.20 . The signal that the single axis cannot recover is precisely the part that lives off-axis, distributed across many directions.

We read this as a statement about *dimensionality*. When the model knows a fact, its truth content is concentrated and a single direction captures most of it. When the model does not know a fact, the truth-relevant signal is weak and diffuse, and one direction recovers only a fraction. The dimensionality of the truth representation is thus not fixed but *knowledge-dependent*. This is, to our knowledge, a framing that neither the linear-probe nor the dictionary-learning literature states in these terms, and it is the contribution we consider most novel.

4.3 The complex plane does not help; “phase” is the sign flip

On single-polarity data the second dimension adds nothing: at layer 16 the 2D logistic (0.931) does not exceed the 1D axis (0.938). This is structural. Minimal pairs cancel the topic, so the differences concentrate along one direction; the SVD places the entire signal in v_1 , and v_2 is residual variance—noise, not a second truth factor. A controlled synthetic check confirms that the apparent “rotation” of falsehoods is an artifact: with a pure sign flip and zero genuine rotation, an angular (phase) score still separates classes, because a state at $\text{Re} < 0$, $\text{Im} \approx 0$ has phase $\approx \pi$ by definition. The earlier intuition of truth-as-phase, in the manner of a critical line, has no operational content here: it reduces to the sign of the real axis re-expressed in polar coordinates. We therefore retract that framing.

4.4 Unsupervised recovery of the polarity direction

A genuine second dimension does exist, but only when the data span both polarities, and it is the polarity direction of [5], not a phase. Table 2 shows the diagnostic: an axis fit on affirmative

Table 2: Polarity analysis (leave-facts-out CV), layer 14 chosen on affirmative data; and unsupervised t_P recovery at layer 15. AUC below 0.5 indicates a sign flip.

Quantity	Value
$\cos(t_A, t_N)$	-0.415
Affirmative axis on affirmative (held-out AUC)	0.981
Affirmative axis on <i>negated</i> (held-out AUC)	0.079
General direction t_G on negated (held-out AUC)	0.954
Unsupervised SVD vs. supervised t_P : $\max_k \cos(\text{SVD}_k, t_P) $	0.959
(principal component carrying t_P)	PC 1
$\cos(\text{SVD}_1, t_G)$	0.157

statements classifies affirmative held-out facts at 0.981 but *anti-classifies* negated ones at 0.079—a near-total flip—while the general direction t_G recovers the negated facts at 0.954. The truth direction genuinely rotates with polarity ($\cos(t_A, t_N) = -0.415$).

The methodological result is the recovery. When we run the unsupervised difference SVD on *mixed-polarity* minimal pairs—blind to the negation label—its leading direction aligns with the supervised polarity direction t_P at cosine 0.959 (layer 15), and t_P lands in the first principal component, while alignment with t_G is only 0.157. In other words, when both polarities are present, the dominant axis of the differences *is* polarity, and the SVD finds it without supervision. Where [5] constructs t_P by separating affirmative from negated statements by hand, the same direction emerges here from geometry alone.

5 Discussion: what a single axis can and cannot read

It is tempting to view a cheap geometric probe as a low-cost version of dictionary learning—reading superposition without expanding it. We argue this is impossible, and that the impossibility is a capacity statement, not an implementation gap. A projection is an inner product: it collapses a d -dimensional state to a single scalar, discarding $d - 1$ dimensions. From that scalar one cannot reconstruct which features were present, only how much a chosen direction was. “Reading the superposition” requires asking the question once per dictionary atom, which over an overcomplete dictionary of size m is exactly the $O(dm)$ sparse autoencoder. Compression (project to a scalar) and disambiguation (expand to a sparse code) are not inverse cheap-vs-expensive versions of one operation; they are different operations, and only the second reads superposition. The ≈ 0.20 probe gap of Sec. 4.2 is a direct measurement of the truth information that a single axis structurally cannot return.

The consequence is a clear scope. This method’s value is *structural and cheap*: a training-free axis that, on facts the model knows, rivals a trained probe at two dot products per token; a quantitative law relating the readability of truth to the model’s knowledge of the fact; and an unsupervised route to the polarity direction. It is not a substitute for SAEs and should not be presented as one.

6 Limitations

These results are preliminary. They are measured on a single model (Qwen2.5-1.5B); replication across architectures and scales is required, particularly for the depth of the peak and the knowledge-dependent dimensionality law. The datasets are small (tens to a few hundred pairs), so confidence

intervals are wide despite the permutation control. The “unknown-fact” interpretation of the CounterFact gap is plausible but confounded: a fact the model does not know and a fact whose truth is genuinely multidimensional are not separated here, and disentangling them (e.g. by filtering CounterFact to facts the model can answer) is necessary to make the dimensionality claim quantitative. We use last-token pooling; other aggregations may shift the axis.

7 Conclusion

Stripped of a decorative complex-plane framing, three results survive on a real model under a falsifiable protocol: an unsupervised truth axis carries a permutation-tested signal at cost $O(d)$; the gap between that axis and a full probe scales with how well the model knows the fact, suggesting a knowledge-dependent dimensionality of truth; and the unsupervised difference SVD recovers the supervised polarity direction of [5] at cosine 0.959. Natural next steps are a multi-model study of the dimensionality law, a knowledge-controlled dataset to separate ignorance from genuine multidimensionality, and a benchmarked evaluation of the polar-gate monitor. The aim is a light, readable, almost-free structural probe to sit alongside generation—honest about the boundary it cannot cross.

8 Reproducibility

Every number in this paper is reproduced by a single script, `truth_probe.py`, run on Qwen2.5-1.5B. Each result is an *isolated* subcommand: running one task loads, extracts, and computes only its own data, and touches nothing belonging to the others. A reviewer who wants to verify one figure runs exactly one command.

Setup.

```
pip install torch transformers datasets
```

A CUDA GPU is optional; on CPU the script runs in float32 (the numbers reported here were obtained in float32). No GPU is required to reproduce any table.

Commands (one per result).

```
# Table 1, top row (clean minimal pairs): held-out AUC, ablation,
#                               full-probe baseline, permutation p-value
python truth_probe.py signal --dataset builtin --with-2d --baseline --perm 200
```

```
# Table 1, bottom row (CounterFact / TruthfulQA mix)
python truth_probe.py signal --dataset mix --max-pairs 250 \
    --with-2d --baseline --perm 200
```

```
# Table 2, top (affirmative -> negated flip; the 2D polarity structure)
python truth_probe.py polarity
```

```
# Table 2, bottom (unsupervised SVD recovers the polarity direction t_P)
python truth_probe.py recovery
```

Each subcommand prints a per-layer table and a final verdict block; the peak-layer line is the value quoted in the paper. Add `-h` to any subcommand for its options (e.g. `python truth_probe.py signal -h`).

Data provenance and safety. The clean minimal pairs and the 2×2 polarity set are embedded in the script. The external datasets are loaded as Parquet only and never with remote code: the script passes no `trust_remote_code`, and the model is loaded with safetensors only. For bit-exact reproduction one may pin a dataset commit, e.g. `-rev-counterfact <sha> -rev-truthfulqa <sha>`, or load from a local file with `-file-counterfact <path>`. The exact commit hashes can be read with `HfApi().dataset_info(repo).sha`.

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